128K X 8 BIT LOW POWER CMOS SRAM

FEATURES

- Access time :55ns
- Low power consumption:
  Operating current:10 mA (TYP.)
  Standby current: 1 µA (TYP.)
- Single 2.7V ~ 5.5V power supply
- Fully Compatible with all Competitors 5V product
- Fully Compatible with all Competitors 3.3V product

- Fully static operation
- Tri-state output
- Data retention voltage : 1.5V (MIN.)
- All products are ROHS Compliant
- Package : 32-pin 450 mil SOP
  32-pin 600 mil P-DIP
  32-pin 8mm x 20mm TSOP-I
  32-pin 8mm x 13.4mm sTSOP
  36-ball 6mm x 8mm TFBGA

GENERAL DESCRIPTION

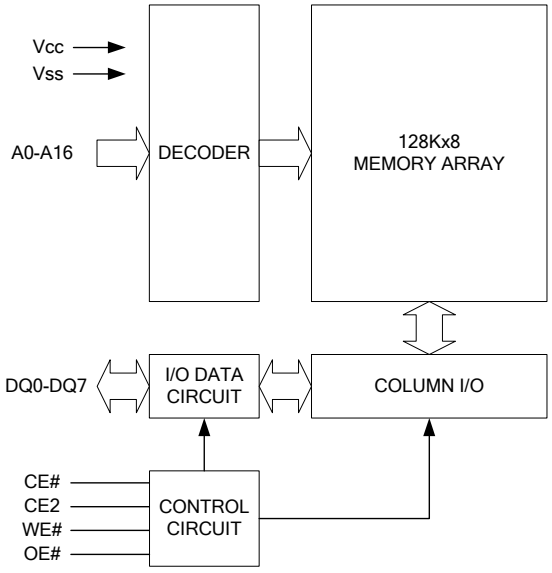
The AS6C1008 is a 1,048,576-bit low power CMOS static random access memory organized as 131,072 words by 8 bits. It is fabricated using very high performance, high reliability CMOS technology. Its standby current is stable within the range of operating temperature.

The AS6C1008 is well designed for very low power system applications, and particularly well suited for battery back-up non-volatile memory application.

The AS6C1008 operates from a single power supply of 2.7V ~ 5.5V.

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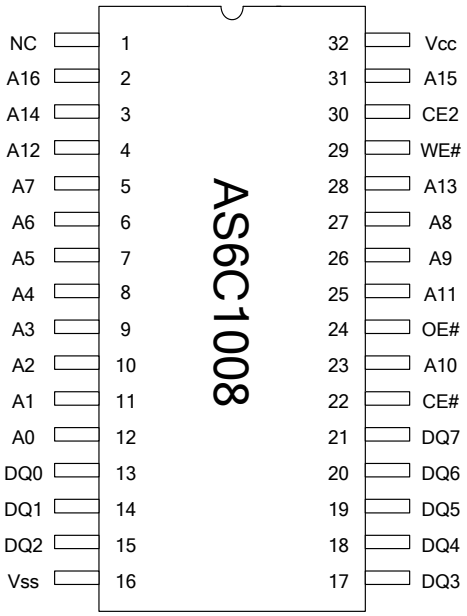
FUNCTIONAL BLOCK DIAGRAM



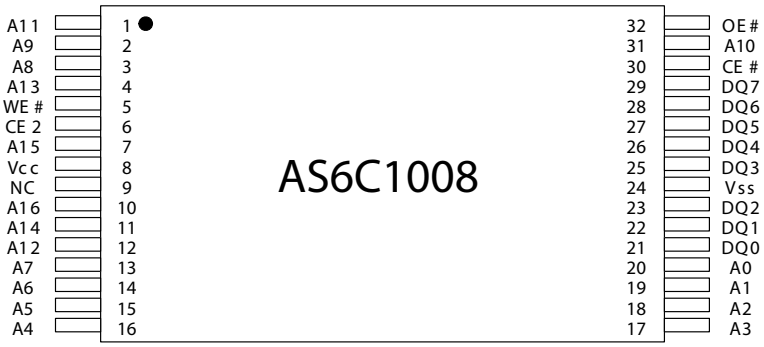
PIN DESCRIPTION

SYMBOL	DESCRIPTION
A0 - A16	Address Inputs
DQ0 – DQ7	Data Inputs/Outputs
CE#, CE2	Chip Enable Inputs
WE#	Write Enable Input
OE#	Output Enable Input
Vcc	Power Supply
Vss	Ground
NC	No Connection

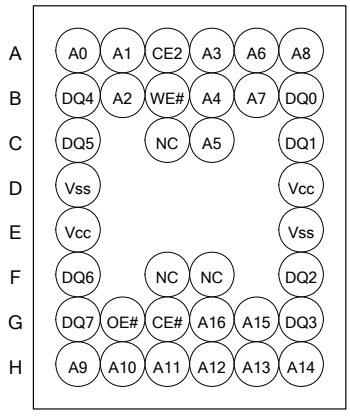
PIN CONFIGURATION



SOP/P-DIP



TSOP-I/sTSOP



TFBGA

ABSOLUTE MAXIMUM RATINGS*

PARAMETER	SYMBOL	RATING	UNIT
Terminal Voltage with Respect to V_{SS}	V_{TERM}	-0.5 to 7.0	V
Operating Temperature	T_A	0 to 70(C grade) -40 to 85(I grade)	ºC
Storage Temperature	T_{STG}	-65 to 150	ºC
Power Dissipation	P_D	1	W
DC Output Current	I_{OUT}	50	mA
Soldering Temperature (under 10 sec)	T_{SOLDER}	260	ºC

*Stresses greater than those listed under "Absolute Maximum Ratings" may cause permanent damage to the device. This is a stress rating only and functional operation of the device or any other conditions above those indicated in the operational sections of this specification is not implied. Exposure to the absolute maximum rating conditions for extended period may affect device reliability.

TRUTH TABLE

MODE	CE#	CE2	OE#	WE#	I/O OPERATION	SUPPLY CURRENT
Standby	H	X	X	X	High-Z	I_{SB1}
	X	L	X	X	High-Z	I_{SB1}
Output Disable	L	H	H	H	High-Z	I_{CC},I_{CC1}
Read	L	H	L	H	D_{OUT}	I_{CC},I_{CC1}
Write	L	H	X	L	D_{IN}	I_{CC},I_{CC1}

Note: H = V_{IH}, L = V_{IL}, X = Don't care.

DC ELECTRICAL CHARACTERISTICS

PARAMETER	SYMBOL	TEST CONDITION		MIN.	TYP.*4	MAX.	UNIT
Supply Voltage	V_{CC}			2.7	3.0	5.5	V
Input High Voltage	V_{IH}*1			0.7*Vcc	-	V_{CC}+0.3	V
Input Low Voltage	V_{IL}*2			- 0.2	-	0.6	V
Input Leakage Current	I_{LI}	$V_{CC} \geqq V_{IN} \geqq V_{SS}$		- 1	-	1	µA
Output Leakage Current	I_{LO}	$V_{CC} \geqq V_{OUT} \geqq V_{SS}$, Output Disabled		- 1	-	1	µA
Output High Voltage	V_{OH}	I_{OH} = -1mA		2.2	2.7	-	V
Output Low Voltage	V_{OL}	I_{OL} = 2mA		-	-	0.4	V
Average Operating Power supply Current	I_{CC}	Cycle time = Min. CE# = V_{IL} and CE2 = V_{IH}, $I_{I/O}$ = 0mA	- 55	-	10	60	mA
	I_{CC1}	Cycle time = 1µs CE#≦0.2V and CE2≧V_{CC}-0.2V, $I_{I/O}$ = 0mA other pins at 0.2V or V_{CC}-0.2V		-	1	10	mA
Standby Power Supply Current	I_{SB1}	CE# ≧V_{CC}-0.2V or CE2≦0.2V	C*	-	1	20	µA
			I*	-	1	50	µA

*C=Commercial temperature/I= Industrial temperature

Notes:
1. V_{IH}(max) = V_{CC} + 3.0V for pulse width less than 10ns.
2. V_{IL}(min) = V_{SS} - 3.0V for pulse width less than 10ns.
3. Over/Undershoot specifications are characterized, not 100% tested.
4. Typical values are included for reference only and are not guaranteed or tested.
   Typical valued are measured at V_{CC} = V_{CC}(TYP.) and T_A = 25ºC

CAPACITANCE (T_A = 25℃, f = 1.0MHz)

PARAMETER	SYMBOL	MIN.	MAX	UNIT
Input Capacitance	C_{IN}	-	6	pF
Input/Output Capacitance	$C_{I/O}$	-	8	pF

Note : These parameters are guaranteed by device characterization, but not production tested.

AC TEST CONDITIONS

Input Pulse Levels	0.2V to V_{CC} - 0.2V
Input Rise and Fall Times	3ns
Input and Output Timing Reference Levels	1.5V
Output Load	C_L =30pF + 1TTL, I_{OH}/I_{OL} = -1mA/2mA

AC ELECTRICAL CHARACTERISTICS

(1) READ CYCLE

PARAMETER	SYM.			AS6C1008-55				UNIT
				MIN.	MAX.			
Read Cycle Time	t_{RC}			55	-			ns
Address Access Time	t_{AA}			-	55			ns
Chip Enable Access Time	t_{ACE}			-	55			ns
Output Enable Access Time	t_{OE}			-	30			ns
Chip Enable to Output in Low-Z	t_{CLZ}*			10	-			ns
Output Enable to Output in Low-Z	t_{OLZ}*			5	-			ns
Chip Disable to Output in High-Z	t_{CHZ}*			-	20			ns
Output Disable to Output in High-Z	t_{OHZ}*			-	20			ns
Output Hold from Address Change	t_{OH}			10	-			ns

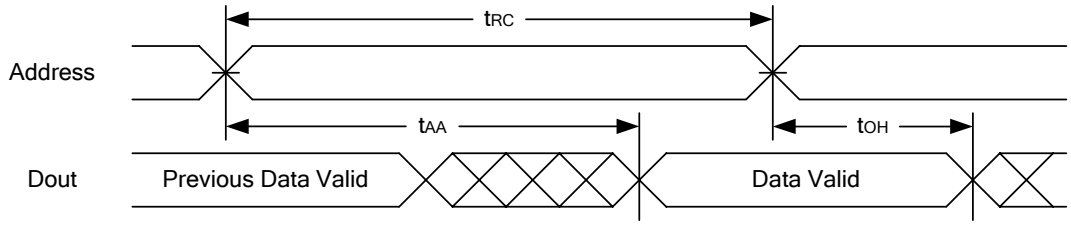
(2) WRITE CYCLE

PARAMETER	SYM.			AS6C1008-55				UNIT
				MIN.	MAX.			
Write Cycle Time	t_{WC}			55	-			ns
Address Valid to End of Write	t_{AW}			50	-			ns
Chip Enable to End of Write	t_{CW}			50	-			ns
Address Set-up Time	t_{AS}			0	-			ns
Write Pulse Width	t_{WP}			45	-			ns
Write Recovery Time	t_{WR}			0	-			ns
Data to Write Time Overlap	t_{DW}			25	-			ns
Data Hold from End of Write Time	t_{DH}			0	-			ns
Output Active from End of Write	t_{OW}*			5	-			ns
Write to Output in High-Z	t_{WHZ}*			-	20			ns

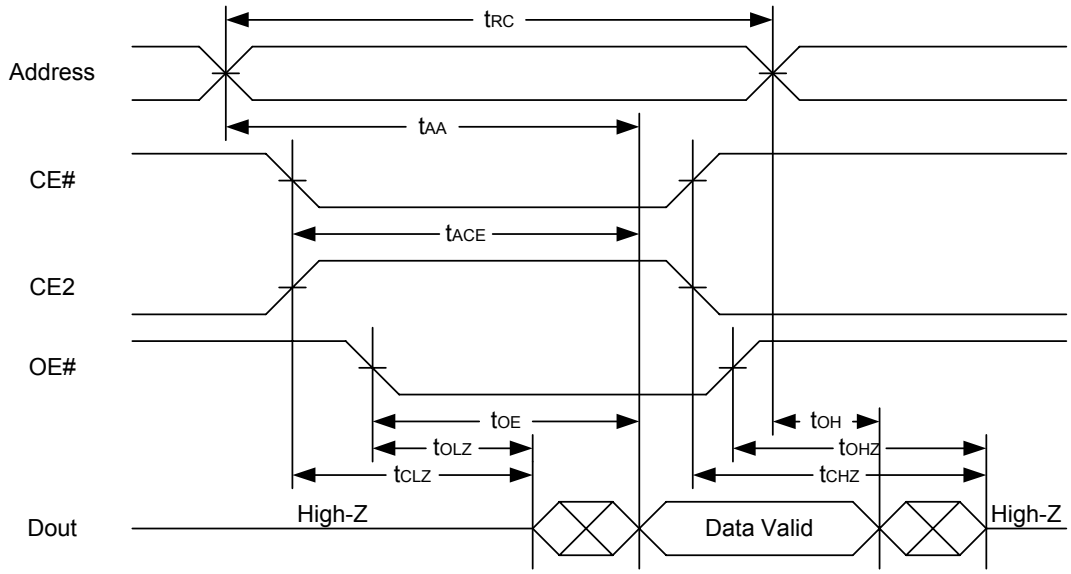
*These parameters are guaranteed by device characterization, but not production tested.

TIMING WAVEFORMS

READ CYCLE 1 (Address Controlled) (1,2)

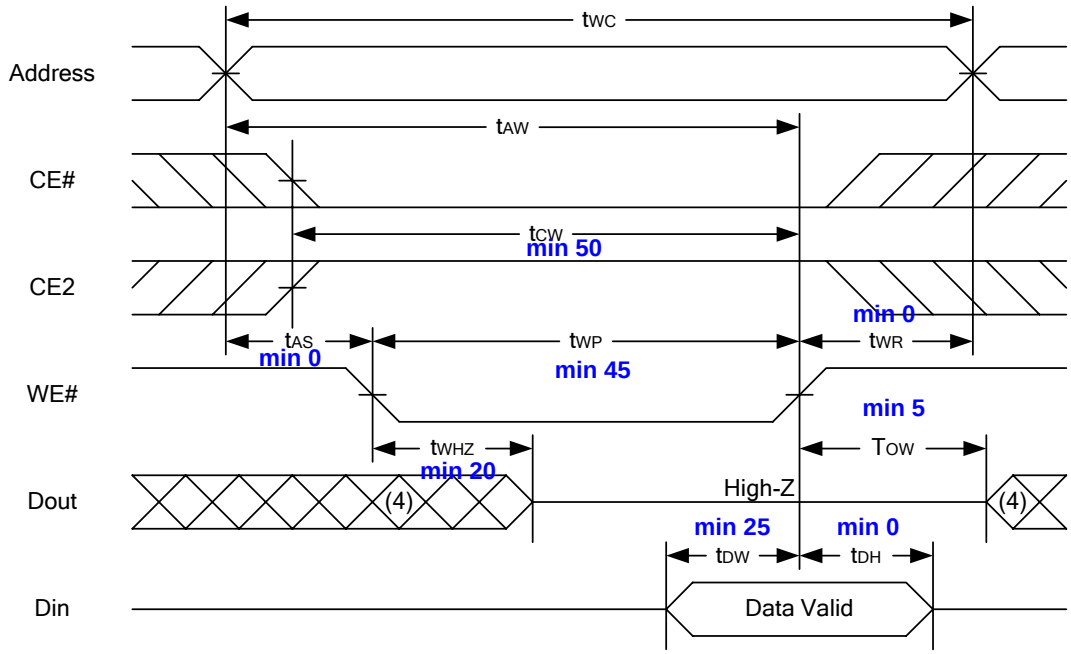


READ CYCLE 2 (CE# and CE2 and OE# Controlled) (1,3,4,5)

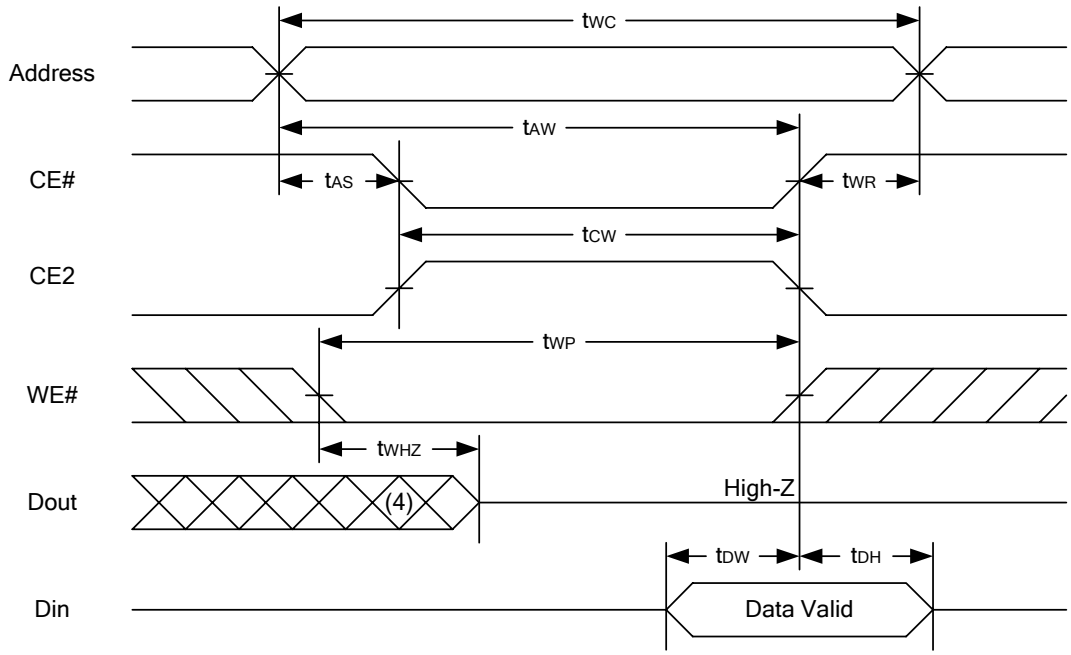


Notes :
1.WE# is high for read cycle.
2.Device is continuously selected OE# = low, CE# = low., CE2 = high.
3.Address must be valid prior to or coincident with CE# = low, CE2 = high; otherwise t_{AA} is the limiting parameter.
4.t_{CLZ}, t_{OLZ}, t_{CHZ} and t_{OHZ} are specified with C_L = 5pF. Transition is measured ±500mV from steady state.
5.At any given temperature and voltage condition, t_{CHZ} is less than t_{CLZ} , t_{OHZ} is less than t_{OLZ}.

WRITE CYCLE 1 (WE# Controlled) (1,2,3,5,6)



WRITE CYCLE 2 (CE# and CE2 Controlled) (1,2,5,6)



Notes :
1.WE#, CE# must be high or CE2 must be low during all address transitions.
2.A write occurs during the overlap of a low CE#, high CE2, low WE#.
3.During a WE#controlled write cycle with OE# low, t_{WP} must be greater than $t_{WHZ} + t_{DW}$ to allow the drivers to turn off and data to be placed on the bus.
4.During this period, I/O pins are in the output state, and input signals must not be applied.
5.If the CE#low transition and CE2 high transition occurs simultaneously with or after WE# low transition, the outputs remain in a high impedance state.
6.t_{OW} and t_{WHZ} are specified with C_L = 5pF. Transition is measured ±500mV from steady state.

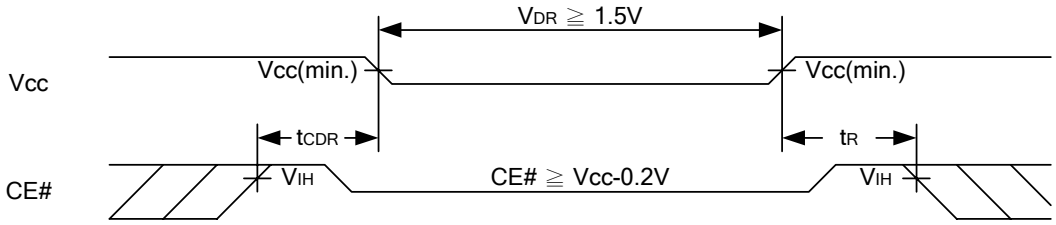
DATA RETENTION CHARACTERISTICS

PARAMETER	SYMBOL	TEST CONDITION		MIN.	TYP.	MAX.	UNIT
V_{CC} for Data Retention	V_{DR}	CE# ≧ V_{CC} - 0.2V or CE2 ≦ 0.2V		1.5	-	5.5	V
Data Retention Current	I_{DR}	V_{CC} = 1.5V CE# ≧ V_{CC} - 0.2V or CE2 ≦ 0.2V	C**	-	0.5	12	µA
			I**		0	30	µA
Chip Disable to Data Retention Time	t_{CDR}	See Data Retention Waveforms (below)		0	-	-	ns
Recovery Time	t_R			t_{RC*}	-	-	ns

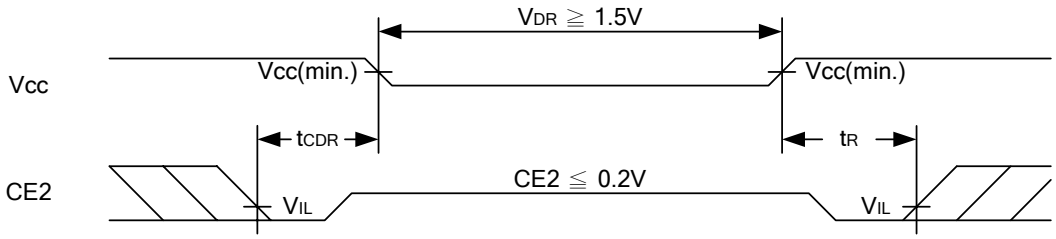
t_{RC*} = Read Cycle Time C=Commercial temp/I = Industrial temp**

DATA RETENTION WAVEFORM

Low Vcc Data Retention Waveform (1) (CE# controlled)

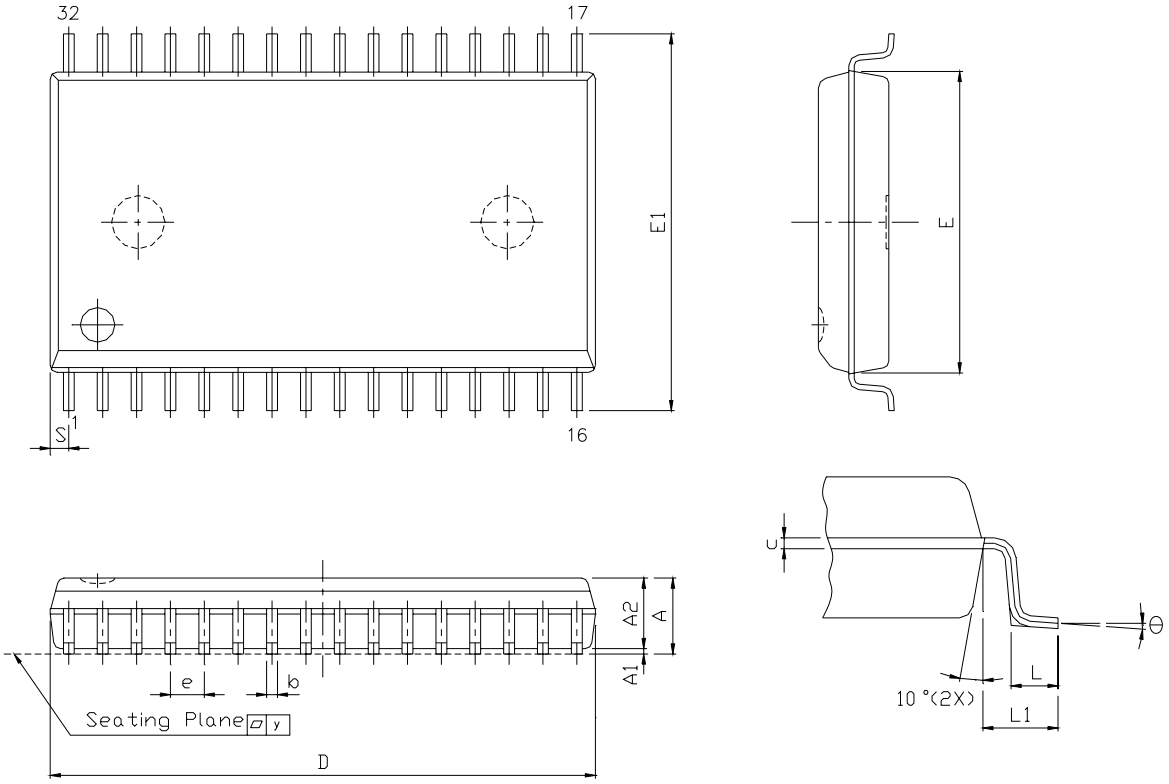


Low Vcc Data Retention Waveform (2) (CE2 controlled)



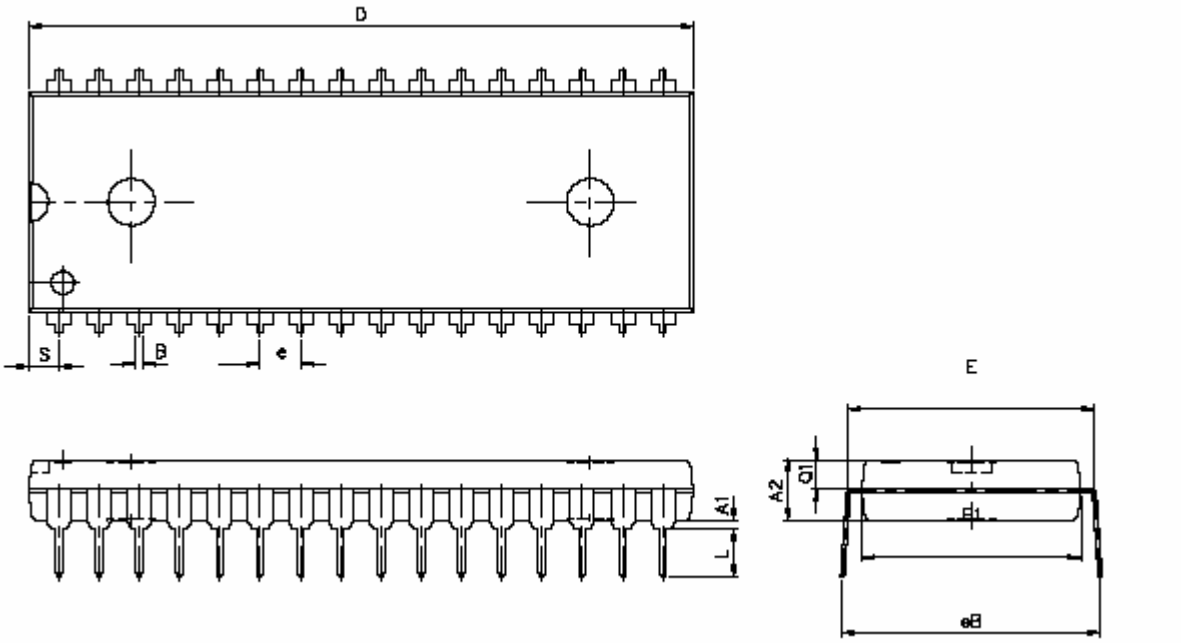
PACKAGE OUTLINE DIMENSION

32 pin 450 mil SOP Package Outline Dimension



UNIT / SYM.	INCH.(BASE)	MM(REF)
A	0.118 (MAX)	2.997 (MAX)
A1	0.004(MIN)	0.102(MIN)
A2	0.111(MAX)	2.82(MAX)
b	0.016(TYP)	0.406(TYP)
c	0.008(TYP)	0.203(TYP)
D	0.817(MAX)	20.75(MAX)
E	0.445 ±0.005	11.303 ±0.127
E1	0.555 ±0.012	14.097 ±0.305
e	0.050(TYP)	1.270(TYP)
L	0.0347 ±0.008	0.881 ±0.203
L1	0.055 ±0.008	1.397 ±0.203
S	0.026(MAX)	0.660 (MAX)
y	0.004(MAX)	0.101(MAX)
Θ	0° -10°	0° -10°

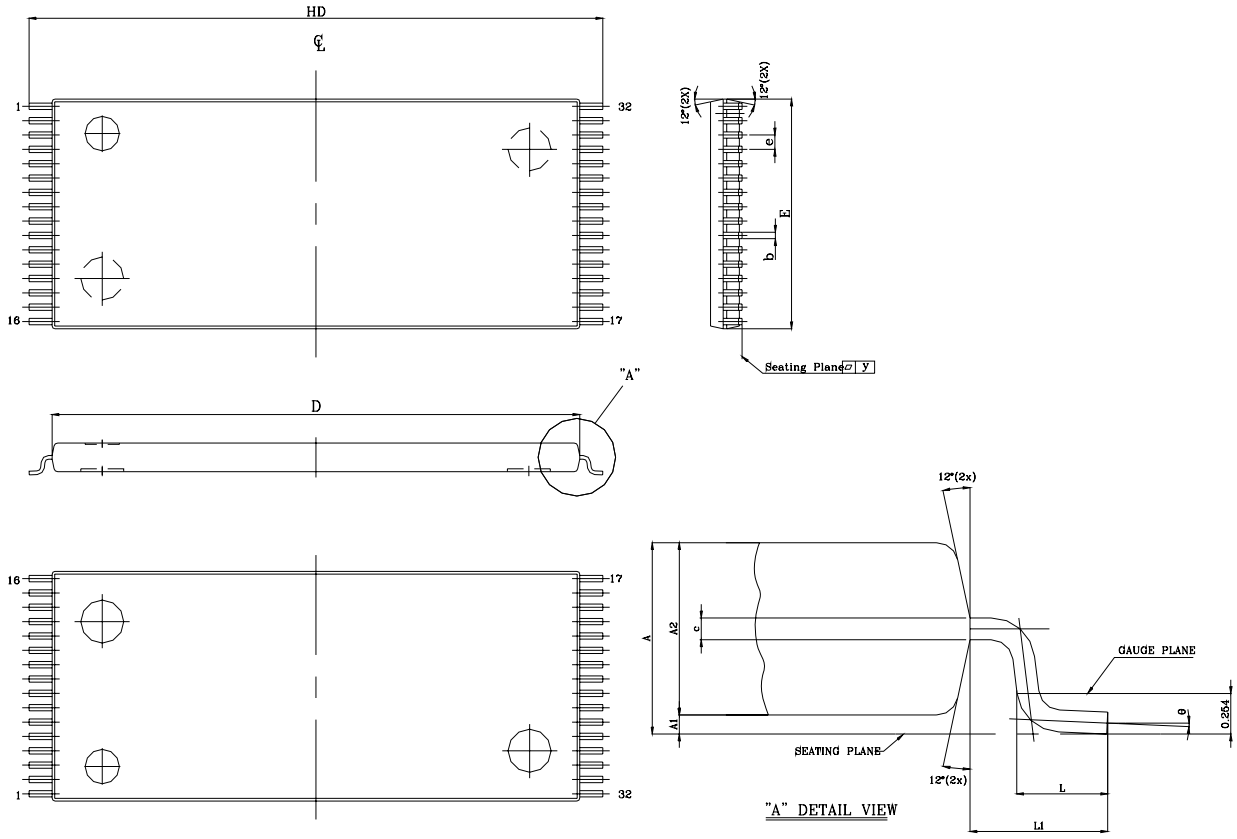
32 pin 600 mil P-DIP Package Outline Dimension



UNIT / SYM.	INCH(BASE)	MM(REF)
A1	0.001 (MIN)	0.254 (MIN)
A2	0.150 ± 0.005	3.810 ± 0.127
B	0.018 ± 0.005	0.457 ± 0.127
D	1.650 ± 0.005	41.910 ± 0.127
E	0.600 ± 0.010	15.240 ± 0.254
E1	0.544 ± 0.004	13.818 ± 0.102
e	0.100 (TYP)	2.540 (TYP)
eB	0.640 ± 0.020	16.256 ± 0.508.
L	0.130 ± 0.010	3.302 ± 0.254
S	0.075 ± 0.010	1.905 ± 0.254
Q1	0.070 ± 0.005	1.778 ± 0.127

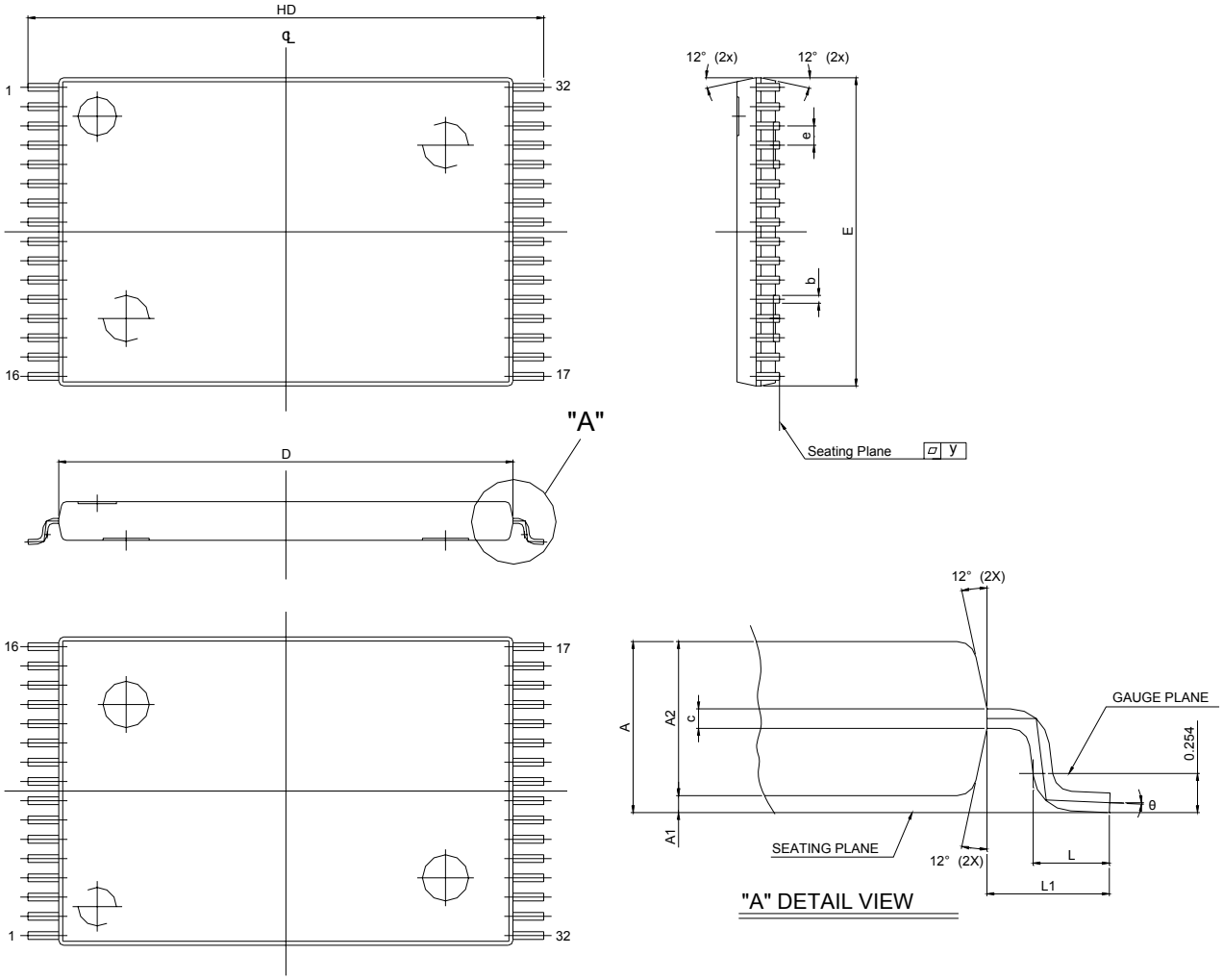
Note : D/E1/S dimension do not include mold flash.

32 pin 8mm x 20mm TSOP-I Package Outline Dimension



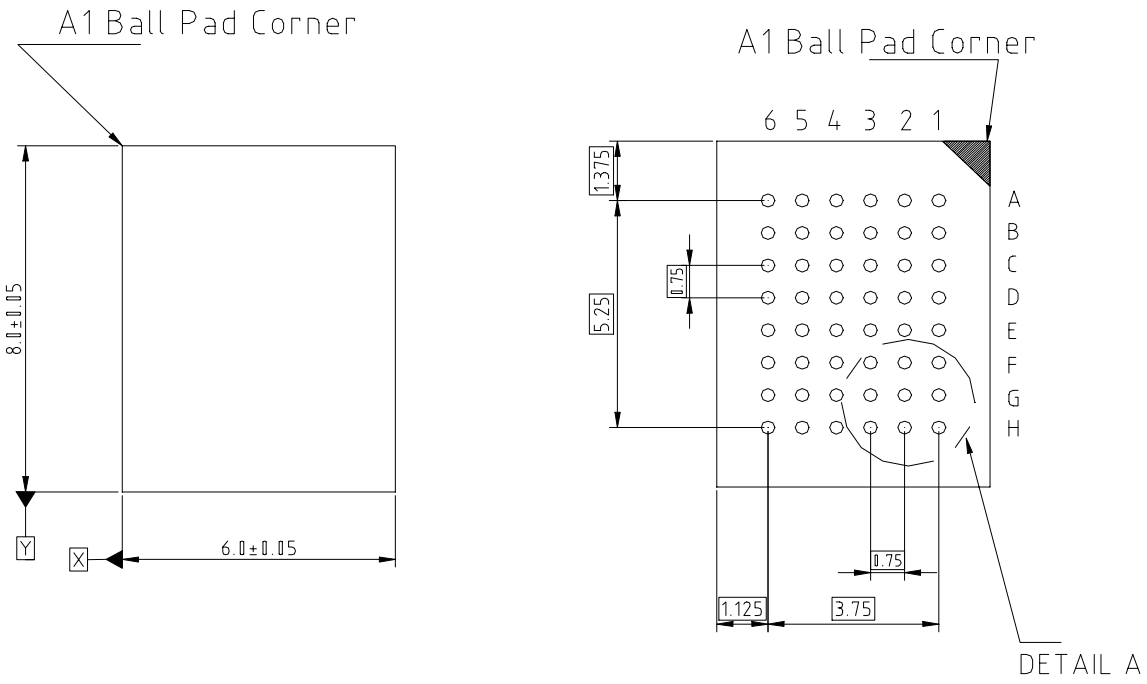
UNIT / SYM.	INCH(BASE)	MM(REF)
A	0.047 (MAX)	1.20 (MAX)
A1	0.004 ±0.002	0.10 ±0.05
A2	0.039 ±0.002	1.00 ±0.05
b	0.008 + 0.002 - 0.001	0.20 + 0.05 -0.03
c	0.005 (TYP)	0.127 (TYP)
D	0.724 ±0.004	18.40 ±0.10
E	0.315 ±0.004	8.00 ±0.10
e	0.020 (TYP)	0.50 (TYP)
HD	0.787 ±0.008	20.00 ±0.20
L	0.0197 ±0.004	0.50 ±0.10
L1	0.0315 ±0.004	0.08 ±0.10
y	0.003 (MAX)	0.076 (MAX)
Θ	0°～5°	0°～5°

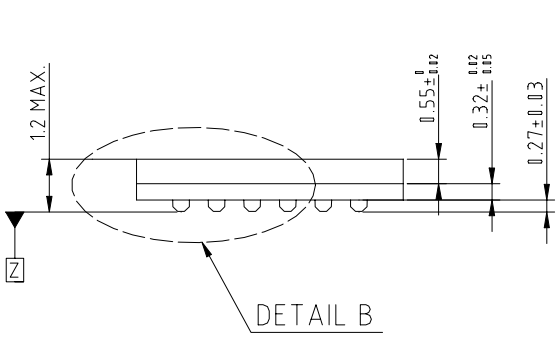
32 pin 8mm x 13.4mm sTSOP Package Outline Dimension



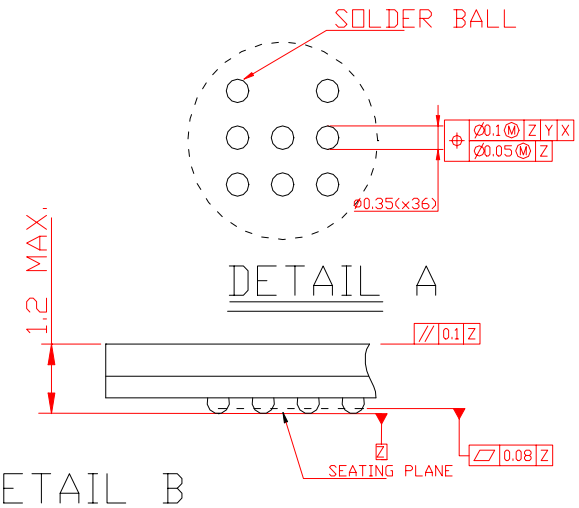
SYM. \ UNIT	INCH(BASE)	MM(REF)
A	0.049 (MAX)	1.25 (MAX)
A1	0.005 ±0.002	0.130 ±0.05
A2	0.039 ±0.002	1.00 ±0.05
b	0.008 ±0.01	0.20±0.025
c	0.005 (TYP)	0.127 (TYP)
D	0.465 ±0.004	11.80 ±0.10
E	0.315 ±0.004	8.00 ±0.10
e	0.020 (TYP)	0.50 (TYP)
HD	0.528±0.008	13.40 ±0.20.
L	0.0197 ±0.004	0.50 ±0.10
L1	0.0315 ±0.004	0.8 ±0.10
y	0.003 (MAX)	0.076 (MAX)
Θ	0°～5°	0°～5°

36 ball 6mm × 8mm TFBGA Package Outline Dimension





SIDE VIEW



DETAIL B

ORDERING INFORMATION

Ordering Codes

Alliance	Organization	VCC range	Package	Operating Temp	Speed ns
AS6C1008-55PCN	128K X 8	2.7-5.5V	32pin 600mil PDIP	Commercial ~ 0° C to 70° C	55
AS6C1008-55PIN	128K X 8	2.7-5.5V	32pin 600mil PDIP	Industrial ~ -40°C to 85° C	55
AS6C1008-55SIN	128K X 8	2.7-5.5V	32pin 450mil SOP	Industrial ~ -40°C to 85° C	55
AS6C1008-55TIN	128K X 8	2.7-5.5V	32pin TSOP-I (8 x 20 mm)	Industrial ~ -40°C to 85° C	55
AS6C1008-55STIN	128K X 8	2.7-5.5V	32pin sTSOP (8 x 13.4 mm)	Industrial ~ -40°C to 85° C	55
AS6C1008-55BIN	128K X 8	2.7-5.5V	36pin TFBGA (6mm x 8mm)	Industrial ~ -40°C to 85° C	55

Part numbering system

AS6C	1008	- 55	X	X	N
low power SRAM prefix	Device Number 10 = 1M 08 = by 8	Access Time	Package Options: P = 32 pin 600 mil P-DIP S = 32 pin 450 mil SOP T = 32 pin TSOP-I (8mm x 20 mm) ST = 32 pin sTSOP (8 x 13.4 mm) B = 36 ball 6 x 8mm TFBGA	Temperature Range: C = Commercial (0°C to +70° C) I = Industrial (-40° to +85° C)	N = Lead Free ROHS Compliant Part

128K X 8 BIT LOW POWER CMOS SRAM

Alliance Memory, Inc.
1116 South Amphlett, #2,
San Mateo, CA 94402
Tel: 650-525-3737
Fax: 650-525-0449
www.alliancememory.com

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Part Number: AS6C1008
Document Version: v. 1.0

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